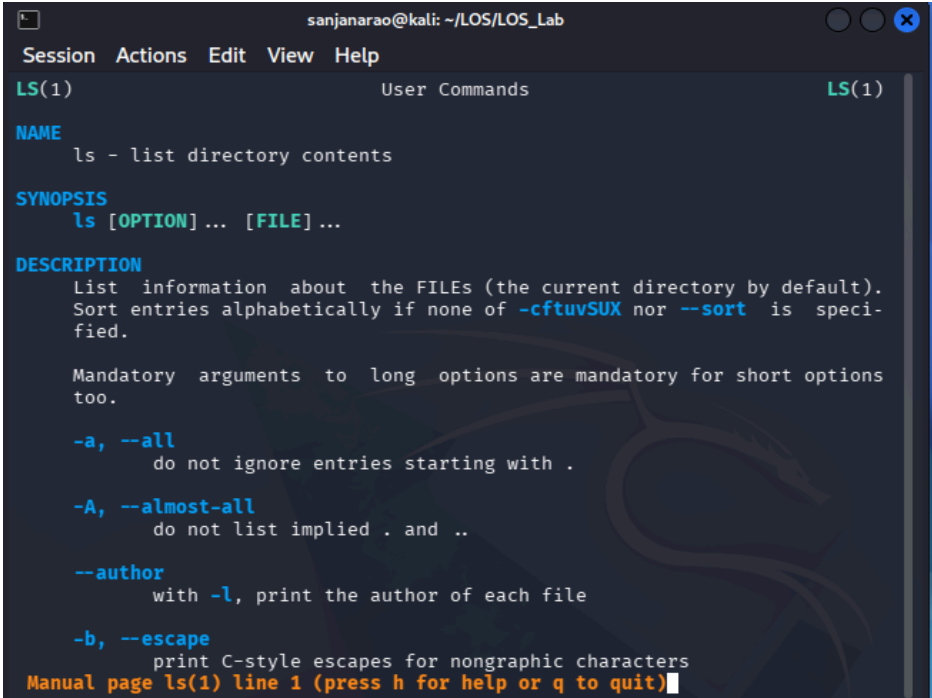


## LOS\_Lab Assignment 2

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The main aim of this lab session is to provide hands-on experience on:

### 1. Getting Help

| Task                                                                                | Command Name | Syntax        | Example      |
|-------------------------------------------------------------------------------------|--------------|---------------|--------------|
| To get manual page for the known command                                            | man          | man <command> | man ls       |
|  |              |               |              |
| To get manual page for the unknown command                                          | apropos      | apropos       | apropos copy |

```

sanjanarao@kali: ~/LOS/LOS_Lab
Session Actions Edit View Help
scp (1) - OpenSSH secure file copy
scp1 (1) - secure copy (remote file copy program)
ssh-copy-id (1) - use locally available keys to authorise logins on a ...
stpcpy (3) - string return-offset-pointer copy
strcat (3) - copy or concatenate a string
strcpy (3) - copy or concatenate a string
string_copying (7) - copying strings and character sequences
svnversion (1) - Produce a compact version identifier for a working ...
UFFDIO_COPY (2const) - atomically copy a continuous memory chunk into the ...
uuid_copy (3) - copy a UUID value
va_copy (3) - variable argument lists
vfs_shadow_copy (8) - Expose snapshots to Windows clients as shadow copies.
vfs_shadow_copy2 (8) - Expose snapshots to Windows clients as shadow copies.
wcpcpy (3) - copy a wide-character string, returning a pointer t...
wcpncpy (3) - copy a fixed-size string of wide characters, return...
wcscpy (3) - copy a wide-character string
wcsncpy (3) - copy a fixed-size string of wide characters
wmemcpy (3) - copy an array of wide-characters
wmemmove (3) - copy an array of wide-characters
wmempcpy (3) - copy memory area
x86_64-linux-gnu-objcopy (1) - copy and translate object files
x86_64-w64-mingw32-objcopy (1) - copy and translate object files
xclip-copyfile (1) - copy and move files via the X clipboard
xclip-cutfile (1) - copy and move files via the X clipboard
xclip-pastefile (1) - copy and move files via the X clipboard

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$

```

|                                                                                                          |         |                   |            |
|----------------------------------------------------------------------------------------------------------|---------|-------------------|------------|
| To know the source file binary                                                                           | whereis | whereis <command> | whereis ls |
| <pre> (sanjanarao@kali)-[~/LOS/LOS_Lab] \$ whereis ls ls: /usr/bin/ls /usr/share/man/man1/ls.1.gz </pre> |         |                   |            |
| To know the path of the command                                                                          | which   | which <command>   | which cd   |
| <pre> (sanjanarao@kali)-[~/LOS/LOS_Lab] \$ which cd cd: shell built-in command </pre>                    |         |                   |            |
| To know the command is external or internal                                                              | type    | type<command>     | type cd    |
| <pre> (sanjanarao@kali)-[~/LOS/LOS_Lab] \$ type cd cd is a shell builtin </pre>                          |         |                   |            |
| To get help for the internal command                                                                     | help    | help <command>    | help cd    |

```

sanjanarao@kali: ~/LOS/LOS_Lab
Session Actions Edit View Help
-P use the physical directory structure
without following symbolic links: resolve symbolic link
s in DIR before processing instances of '..'
-e if the -P option is supplied, and the
current working directory cannot be determined succes
sfully, exit with
a non-zero status
-a on systems that support it, present a
file with extended
attributes as a directory containing
the file attributes

The default is to follow symbolic links, as if '-
L' were specified.
'..' is processed by removing the immediately pre
vious pathname component
back to a slash or the beginning of DIR.

Exit Status:
Returns 0 if the directory is changed, and if $PW
D is set successfully when
-P is used; non-zero otherwise.

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$

```

To list out bash commands

compugen

compugen - c

compugen -c

```

sanjanarao@kali: ~/LOS/LOS_Lab
Session Actions Edit View Help
rpmkeys
invproj
cat
gcc-nm
redis-check-aof
xdg-user-dirs-gtk-update
grops
socat-broker.sh
tsk_imageinfo
rgb3toppm
avahi-publish
atk6-fake_mipv6
rqworker
gftopk
ldapcompare
proxchains4
httpclient
inetutils-telnet
snmpinform
mf
pamstretch-gen
pc1toppm
fakeroot-tcp
autoexpect
tkconch3

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$

```

To know the usage of the command

whatis

whatis <command>

whatis ls

```

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ whatis ls
ls (1) - list directory contents

```

## 2. Basic commands

| Task                                                                                                                                                                                                                                                                     | Command Name | Syntax             | Example      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------------|--------------|
| To know today's date                                                                                                                                                                                                                                                     | date         | date               | date         |
| <pre>(sanjanarao@kali)-[~]<br/>\$ date<br/>Thursday 20 August 2026 02:11:16 PM IST</pre>                                                                                                                                                                                 |              |                    |              |
| To print calendar                                                                                                                                                                                                                                                        | cal          | cal [month] [year] | cal          |
| <pre>sanjanarao@kali: ~<br/>Session Actions Edit View Help<br/>(sanjanarao@kali)-[~]<br/>\$ cal<br/>August 2026<br/>Su Mo Tu We Th Fr Sa<br/>      1<br/> 2  3  4  5  6  7  8<br/> 9 10 11 12 13 14 15<br/>16 17 18 19 20 21 22<br/>23 24 25 26 27 28 29<br/>30 31</pre> |              |                    |              |
| To print kernel version                                                                                                                                                                                                                                                  | uname        | uname -r           | uname -r     |
| <pre>(sanjanarao@kali)-[~]<br/>\$ uname -r<br/>6.19.14+kali-amd64</pre>                                                                                                                                                                                                  |              |                    |              |
| To print default shell                                                                                                                                                                                                                                                   | echo         | echo \$SHELL       | echo \$SHELL |
| <pre>(sanjanarao@kali)-[~]<br/>\$ echo \$SHELL<br/>/usr/bin/zsh</pre>                                                                                                                                                                                                    |              |                    |              |
| To print currently logged in user                                                                                                                                                                                                                                        | whoami       | whoami             | whoami       |
| <pre>(sanjanarao@kali)-[~]<br/>\$ whoami<br/>sanjanarao</pre>                                                                                                                                                                                                            |              |                    |              |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |                      |                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------------|-----------------------|
| To create shortcut for command                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | alias   | alias name='command' | alias l='ls -l'       |
| <pre>(sanjanarao@kali)-[~] \$ alias l='ls -l'  (sanjanarao@kali)-[~] \$ l total 40 drwxr-xr-x 2 sanjanarao sanjanarao 4096 Aug 18 22:1 3 Desktop drwxr-xr-x 2 sanjanarao sanjanarao 4096 Aug 18 22:1 3 Documents drwxr-xr-x 2 sanjanarao sanjanarao 4096 Aug 18 22:1 3 Downloads drwxrwxr-x 3 sanjanarao sanjanarao 4096 Aug 19 07:0 3 LOS drwxr-xr-x 2 sanjanarao sanjanarao 4096 Aug 18 22:1 3 Music drwxr-xr-x 2 sanjanarao sanjanarao 4096 Aug 20 07:1 2 Pictures drwxr-xr-x 2 sanjanarao sanjanarao 4096 Aug 18 22:1 3 Projects drwxr-xr-x 2 sanjanarao sanjanarao 4096 Aug 18 22:1 3 Public drwxr-xr-x 2 sanjanarao sanjanarao 4096 Aug 18 22:1 3 Templates drwxr-xr-x 2 sanjanarao sanjanarao 4096 Aug 18 22:1 3 Videos</pre> |         |                      |                       |
| To delete shortcut                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | unalias | unalias name         | unalias l             |
| <pre>(sanjanarao@kali)-[~] \$ unalias l  (sanjanarao@kali)-[~] \$ l l: command not found  (sanjanarao@kali)-[~] \$</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |                      |                       |
| To change the timestamp of the file                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | touch   | touch <file>         | touch -m time2.txt    |
| <pre>(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2] \$ touch -m time2.txt</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |                      |                       |
| To create empty files                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | touch   | touch file1 file2    | touch lab2.txt lab2.c |
| <pre>(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2] \$ touch lab2.txt lab2.c</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |                      |                       |
| To clear the screen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | clear   | clear                | clear                 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |                     |                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|---------------------|
| <pre>(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2] \$</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                     |                     |                     |
| To know disk usage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | du                  | du [options] [path] | du -sh .            |
| <pre>(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2] \$ du -sh . 4.0K .</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                     |                     |                     |
| To know free space in the system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | df                  | df [options]        | df -h               |
| <pre>(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2] \$ df -h Filesystem      Size  Used Avail Use% Mounted on udev            933M   0    933M   0% /dev tmpfs           194M  1.2M  193M   1% /run /dev/sda1       19G   15G   3.0G  84% / tmpfs           967M  4.0K  967M   1% /dev/shm none            1.0M   0    1.0M   0% /run/credentials/syste md-journald.service tmpfs           967M  952K  966M   1% /tmp none            1.0M   0    1.0M   0% /run/credentials/getty @tty1.service tmpfs           194M  108K  194M   1% /run/user/1000</pre> |                     |                     |                     |
| To know about the linux release                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | cat /etc/os-release | cat /etc/os-release | cat /etc/os-release |
| <pre>(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2] \$ cat /etc/os-release PRETTY_NAME="Kali GNU/Linux Rolling" NAME="Kali GNU/Linux" VERSION_ID="2026.2" VERSION="2026.2" VERSION_CODENAME=kali-rolling ID=kali ID_LIKE=debian HOME_URL="https://www.kali.org/" SUPPORT_URL="https://forums.kali.org/" BUG_REPORT_URL="https://bugs.kali.org/" ANSI_COLOR="1;31"</pre>                                                                                                                                                                                      |                     |                     |                     |
| To show login history                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | last                | last                | last                |

|                                                                                                                                                                                                                                                                                                                                                                            |             |                      |                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------|----------------------|
| <pre> (sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2] \$ last sanjanar tty7      :0                Thu Aug 20 17:59 - st ill logged in sanjanar tty7      :0                Thu Aug 20 16:08 - st ill logged in sanjanar tty7      :0                Thu Aug 20 14:18 - st ill logged in sanjanar tty7      :0                Thu Aug 20 13:56 - st </pre>                          |             |                      |                      |
| To check the current user privileges                                                                                                                                                                                                                                                                                                                                       | sudo        | sudo -l              | sudo -l              |
| <pre> (sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2] \$ sudo -l Matching Defaults entries for sanjanarao on kali:     secure_path=/usr/sbin\::/usr/bin\::/sbin\::/bin  Runas and Command-specific defaults for sanjanarao:     Defaults!/usr/sbin/visudo env_keep+="SUDO_EDITOR EDITOR VISUAL"  User sanjanarao may run the following commands on kali:     (ALL : ALL) ALL </pre> |             |                      |                      |
| To find about the Linux release                                                                                                                                                                                                                                                                                                                                            | lsb_release | lsb_release -a       | lsb_release -a       |
| <pre> (sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2] \$ lsb_release -a No LSB modules are available. Distributor ID: Kali Description:    Kali GNU/Linux Rolling Release:        2026.2 Codename:       kali-rolling </pre>                                                                                                                                                        |             |                      |                      |
| To check system uptime                                                                                                                                                                                                                                                                                                                                                     | uptime      | uptime               | uptime               |
| <pre> (sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2] \$ uptime 19:16:19 up 1:18, 1 user, load average: 0.85, 0.41, 0.22 </pre>                                                                                                                                                                                                                                                     |             |                      |                      |
| To shutdown                                                                                                                                                                                                                                                                                                                                                                | shutdown    | sudo shutdown -h now | sudo shutdown -h now |
| <pre> (sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2] \$ sudo shutdown -h now </pre>                                                                                                                                                                                                                                                                                                |             |                      |                      |

|                                                 |        |             |             |
|-------------------------------------------------|--------|-------------|-------------|
| To restart                                      | reboot | sudo reboot | sudo reboot |
| <pre>(sanjanarao@kali)-[~] \$ sudo reboot</pre> |        |             |             |

### 3. Navigation

| Task                                                                                              | Command | Syntax            | Example    |
|---------------------------------------------------------------------------------------------------|---------|-------------------|------------|
| To navigate home directory                                                                        | cd      | cd ~              | cd ~       |
| <pre>(sanjanarao@kali)-[/home] \$ cd ~</pre>                                                      |         |                   |            |
| To navigate to the parent directory                                                               | cd      | cd ..             | cd ..      |
| <pre>(sanjanarao@kali)-[~] \$ cd .. (sanjanarao@kali)-[/home] \$</pre>                            |         |                   |            |
| To navigate to the child directory                                                                | cd      | cd <directory>    | cd Lab2    |
| <pre>(sanjanarao@kali)-[~/LOS/LOS_Lab] \$ cd Lab2 (sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2] \$</pre> |         |                   |            |
| Alternate command to cd                                                                           | pushd   | pushd <directory> | pushd Lab2 |



|                                                                                                                                          |      |      |      |
|------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|
| <pre> (sanjanarao@kali)-[~/LOS/LOS_Lab] \$ pushd Lab2 ~/LOS/LOS_Lab/Lab2 ~/LOS/LOS_Lab  (sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2] \$ </pre> |      |      |      |
| To go back to the previous directory                                                                                                     | cd - | cd - | cd - |
| <pre> (sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2] \$ cd - ~/LOS/LOS_Lab  (sanjanarao@kali)-[~/LOS/LOS_Lab] \$ </pre>                          |      |      |      |
| To go to the root directory                                                                                                              | cd   | cd / | cd / |
| <pre> (sanjanarao@kali)-[~/LOS/LOS_Lab] \$ cd /  (sanjanarao@kali)-[/] \$ </pre>                                                         |      |      |      |

#### 4. File System

| Task                                                                                                                                                                   | Syntax       | Command |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------|
| How to identify the file system                                                                                                                                        | df -T [path] | df -T . |
| <pre> (sanjanarao@kali)-[~] \$ df -T .  Filesystem      Type 1K-blocks    Used Available Use% Mounted on /dev/sda1      ext4 19353424 15298768   3046224  84% / </pre> |              |         |

## 5. Globbing

Create a dedicated practice directory containing sample files such as:

index.html, login.html, admin.htm, report.txt, access.log, error.log, user1.txt, user2.log, file123, backup1.html, backup2.txt

| No | Task                                                                           | Pattern             |
|----|--------------------------------------------------------------------------------|---------------------|
| a  | List all files ending in .html.                                                | ls *.html           |
| b  | List all files other than .html files using extended globbing.                 | ls !(*.html)        |
| c  | List filenames containing at least one numeric digit.                          | ls *[0-9]*          |
| d  | List files whose names begin with user followed by a single digit.             | ls user[0-9].*      |
| e  | List files ending in either .txt or .log using a single extended glob pattern. | ls +(*.txt *.log)   |
| f  | List files that do not end in .txt or .log.                                    | ls *.{txt,log}      |
| g  | List files beginning with either access or error using extended globbing.      | ls {access*,error*} |
| h  | List .html files whose filenames contain a numeric digit.                      | ls *[0-9]*.html     |

**Result:**

```

(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2/glob_assgin]
$ ls *.html
backup1.html  index.html  login.html

(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2/glob_assgin]
$ ls !(*.html)
access.log  admin.htm  backup2.txt  error.log  file123  report.txt  user1.txt  user2.log

(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2/glob_assgin]
$ ls *[0-9]*
backup1.html  backup2.txt  file123  user1.txt  user2.log

(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2/glob_assgin]
$ ls user[0-9].*
user1.txt  user2.log

(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2/glob_assgin]
$ ls +(*.txt|*.log)
access.log  backup2.txt  error.log  report.txt  user1.txt  user2.log

(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2/glob_assgin]
$ ls *.{txt,log}
access.log  backup2.txt  error.log  report.txt  user1.txt  user2.log

(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2/glob_assgin]
$ ls {access*,error*}
access.log  error.log

(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2/glob_assgin]
$ ls *[0-9]*.html
backup1.html

```

## 6 . Extended Globbing

### 1. Assume a directory contains:

access.log, error.log, auth.log, auth.log.1, auth.log.2, system.log, notes.txt

| No | Task                                                                                                                              | Pattern                             |
|----|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| a  | Write an extended glob pattern that selects only access.log, error.log, and auth.log.                                             | ls @(access.log error.log auth.log) |
| b  | Select all files except notes.txt.                                                                                                | ls !(@(notes.txt))                  |
| c  | Select files named either auth.log.1 or auth.log.2 using pattern matching rather than writing both complete filenames separately. | ls auth.log.@(1 2)                  |

**Result:**

```
(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2/Extended_glob]
$ ls auth.log.@(1|2)
auth.log.1  auth.log.2

(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2/Extended_glob]
$ ls !(@(notes.txt))
access.log  auth.log  auth.log.1  auth.log.2  error.log  system.log

(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2/Extended_glob]
$ ls @(access.log|error.log|auth.log)
access.log  auth.log  error.log
```

**2. Given the files:**

**malware1.log, malware2.log, malwareA.log, malware\_report.txt, normal.log**

| No | Task                                                                                                                | Pattern               |
|----|---------------------------------------------------------------------------------------------------------------------|-----------------------|
| a  | construct a glob pattern that identifies .log files whose names start with malware and contain a numeric character. | ls malware*[0-9]*.log |

**Result:**

```
(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2/Extended_glob]
$ ls malware*[0-9]*.log
malware1.log  malware2.log
```

### 3. Given:

failed\_login.log, successful\_login.log, login\_backup.log, login.txt, firewall.log

| No | Task                                                                                           | Pattern                                    |
|----|------------------------------------------------------------------------------------------------|--------------------------------------------|
| a  | use extended globbing to select .log files containing either failed_login or successful_login. | ls<br>@(failed_login successful_login).log |
| b  | Construct a pattern that lists all .log files except firewall.log.                             | ls !(@(firewall.log))                      |

### Result:

```
(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2/Extended_glob]
$ ls @(failed_login|successful_login).log
failed_login.log  successful_login.log

(sanjanarao@kali)-[~/LOS/LOS_Lab/Lab2/Extended_glob]
$ ls !(@(firewall.log))
access.log  error.log  malware1.log  normal.log
auth.log    failed_login.log  malware2.log  notes.txt
auth.log.1  login_backup.log  malwareA.log  successful_login.log
auth.log.2  login.txt        malware_report.txt  system.log
```

### 7.Filters

| No | Task                                       | Command              |
|----|--------------------------------------------|----------------------|
| 1  | Display the first 10 entries of login.log. | head -n 10 login.log |
| 2  | Display the first 5 entries of access.log  | head -n 5 access.log |

|    |                                                                                        |                           |
|----|----------------------------------------------------------------------------------------|---------------------------|
| 3  | Display the first 15 security events from security.log.                                | head -n 15 security.log   |
| 4  | Display the first 20 events recorded in incident.log.                                  | head -n 20 incident.log   |
| 5  | Display only the first entry in login.log.                                             | head -n 1 login.log       |
| 6  | Display the first 25 web requests recorded in access.log.                              | head -n 25 access.log     |
| 7  | Display the last 10 entries of login.log.                                              | tail -n 10 login.log      |
| 8  | Display the last 5 entries of access.log.                                              | tail -n 5 access.log      |
| 9  | Display the 20 most recent security events from security.log.                          | tail -n 20 security.log   |
| 10 | Display the 25 most recent events from incident.log.                                   | tail -n 25 incident.log   |
| 11 | Display only the last entry in security.log.                                           | tail -n 1 security.log    |
| 12 | Display the last 50 events recorded in incident.log.                                   | tail -n 50 incident.log   |
| 13 | Display the contents of login.log starting from line 41 until the end of the file.     | tail -n +41 login.log     |
| 14 | Display the contents of security.log starting from line 101 until the end of the file. | tail -n +101 security.log |
| 15 | Find the total number of lines in login.log.                                           | wc -l login.log           |

|    |                                                                                                       |                                      |
|----|-------------------------------------------------------------------------------------------------------|--------------------------------------|
| 16 | Find the total number of lines in access.log.                                                         | wc -l access.log                     |
| 17 | Find the total number of security events in security.log.                                             | wc -l security.log                   |
| 18 | Determine the total number of events recorded in incident.log.                                        | wc -l incident.log                   |
| 19 | Count the number of words in login.log.                                                               | wc -w login.log                      |
| 20 | Count the number of bytes in security.log.                                                            | wc -c security.log                   |
| 21 | Display the line, word, and byte counts of access.log.                                                | wc -l -w -c access.log               |
| 22 | Find the line counts of all four .log files using a single wc command.                                | wc -l *.log                          |
| 23 | Display the first 20 entries of login.log and, from those entries, display only the last 5.           | head -n 20 login.log   tail -n 5     |
| 24 | Display the last 20 entries of login.log and, from those entries, display only the first 5.           | tail -n 20 login.log   head -n 5     |
| 25 | Display the first 30 entries of access.log and then display only the last 10 entries from that output | head -n 30 access.log   tail -n 10   |
| 26 | Display the last 30 entries of access.log and then display only the first 10 entries.                 | tail -n 30 access.log   head -n 10   |
| 27 | Display the first 50 events from security.log and then display the last 10 events from those 50.      | head -n 50 security.log   tail -n 10 |
| 28 | Display the first 50 events from security.log and then display the first 10 events from those 50.     | tail -n 50 security.log   head -n 10 |

|    |                                                                    |                                                                                |
|----|--------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 29 | Display entries from line 11 to line 20 of login.log.              | tail -n +11 login.log   head -n 10<br>or<br>sed -n '11,20p' login.log          |
| 30 | Display entries from line 21 to line 30 of login.log.              | tail -n +21 login.log   head -n 10<br>or<br>sed -n '21,30p' login.log          |
| 31 | Display entries from line 11 to line 20 of access.log.             | tail -n +11 access.log   head -n 10<br>or<br>sed -n '11,20p' access.log        |
| 32 | Display entries from line 31 to line 40 of access.log.             | tail -n +31 access.log   head -n 10<br>or<br>sed -n '31,40p' access.log        |
| 33 | Display security events from line 21 to line 30 of security.log.   | tail -n +21 security.log   head -n 10<br>or<br>sed -n '21,30p' security.log    |
| 34 | Display security events from line 51 to line 60 of security.log.   | tail -n +51 security.log   head -n 10<br>or<br>sed -n '51,60p' security.log    |
| 35 | Display incident events from line 101 to line 110 of incident.log. | tail -n +101 incident.log   head -n 10<br>or<br>sed -n '101,110p' incident.log |
| 36 | Display incident events from line 201 to line 220 of incident.log. | tail -n +201 incident.log   head -n 20<br>or<br>sed -n '201,220p' incident.log |
| 37 | Display incident events from line 491 to line 500.                 | tail -n +491 incident.log   head -n 10<br>or<br>sed -n '491,500p' incident.log |



|    |                                                                                                                          |                                                                                                |
|----|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 38 | Take the first 20 entries of login.log and count how many lines are displayed.                                           | head -n 20 login.log   wc -l                                                                   |
| 39 | Take the last 15 entries of access.log and count how many lines are displayed.                                           | tail -n 15 access.log   wc -l                                                                  |
| 40 | Take the first 50 entries of security.log and count the number of words in those entries.                                | head -n 50 security.log   wc -w                                                                |
| 41 | Take the last 25 entries of security.log and count the number of lines.                                                  | tail -n 25 security.log   wc -l                                                                |
| 42 | Extract lines 21–30 from login.log and count the number of extracted lines.                                              | tail -n +21 login.log   head -n 10   wc -l<br>or<br>sed -n '21,30p' login.log   wc -l          |
| 43 | Extract lines 51–60 from access.log and count the number of extracted lines.                                             | tail -n +51 access.log   head -n 10   wc -l<br>or<br>sed -n '51,60p' access.log   wc -l        |
| 44 | Extract events 101–150 from incident.log and count how many events are displayed                                         | tail -n +101 incident.log   head -n 50   wc -l<br>or<br>sed -n '101,150p' incident.log   wc -l |
| 45 | Display the first 100 events from incident.log, take the last 20 events from that output, and count the number of lines. | head -n 100 incident.log   tail -n 20   wc -l                                                  |

